Yasaswi Bhumireddy

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AI & ML Engineer

AI/ML Engineer with 4+ years building production generative AI and multimodal systems across cloud and financial services. Specializes in **LLMs, agentic AI, multi agent workflows, RAG, foundation models, and MCP**. Drives model performance, latency, cost, and reliability across regulated enterprise environments. Coauthor of peer reviewed publications in LLM based education platforms.

Technical Skills

Languages & Backend	Python, SQL, Java, TypeScript, FastAPI, AsyncIO, REST, GraphQL, OpenAPI
GenAI & LLMs	Google Cloud Vertex AI, Gemini Pro, PyTorch, Hugging Face Transformers & Tokenizers, supervised fine tuning, model inference, prompt engineering, prompt caching, LLM evaluation
Agentic AI & Multi Agent	LangGraph, LangChain, LangSmith, multi agent orchestration, tool calling, policy aware routing, Vertex AI Agent Builder, AgentCore, MCP
RAG & Vector Databases	LlamaIndex, OpenSearch, FAISS, Pinecone, Hugging Face sentence transformers, hybrid keyword/vector retrieval, reranking, metadata filters
Guardrails & Evaluation	PII redaction, hallucination detection, content filtering, tool selection benchmarks, grounding validation, MLflow experiment tracking, quality gated releases
Data Engineering	Kafka, PySpark, Airflow, PostgreSQL, Redis, BigQuery, Google Cloud Storage
Cloud & Infrastructure	Google Cloud (Vertex AI, Cloud Run, Pub/Sub, Cloud Monitoring), AWS (S3, SageMaker, Lambda, Bedrock), Docker, Kubernetes, Terraform, CI/CD
Observability	OpenTelemetry, Prometheus, Grafana, CloudWatch, X-Ray, latency, cost, token tracking, drift detection

Professional Experience

Delivery Consultant – Gen AI/ML | Amazon Web Services | Arlington, VA — May 2025 – Present

- Architected **AI self service platform** on Bedrock, AgentCore, and Connect processing 2.1M monthly conversations, reducing repeat contacts 18% and presenting architecture and KPI dashboards to enterprise client leadership.
- Drove benchmarking of **multi turn conversational and tool using agent workflows** with measurable performance and safety metrics, applying prompt caching, batching, and context compaction to reduce overall serving cost 28%.
- Engineered **tiered memory** across AgentCore Memory and S3 for session state and semantic recall with TTL policies, publishing domain APIs as typed MCP tools via AgentCore Gateway and reducing unauthorized API calls 63%.
- Built **scalable pipelines** for training and deploying foundation models on distributed infrastructure, orchestrating LangChain and LlamaIndex with RAG, similarity search, and tool calling, cutting model onboarding from 14 days to 3 days.
- Curated **high quality datasets** with PySpark, fine tuning Nova foundation models on 68K labeled cases to raise accuracy from 84% to 92% with experiment tracking and staged validation gates.
- Deployed **versioned agents** on AgentCore Runtime via CDK and CodePipeline with canary rollout and automated rollback, linking CloudWatch, X-Ray, and OpenTelemetry to model and tool KPIs, cutting triage time 41%.

AI Engineer | Citibank | Risk & Compliance Technology | New York, NY — Jul 2024 – Apr 2025

- Deployed **LLM powered compliance system** on Vertex AI/Gemini Pro to flag suspicious transactions, screen KYC profiles, and draft SARs, cutting case time from 34 to 13 minutes and lifting analyst throughput 2.6×.
- Orchestrated **multi agent workflows** with LangGraph and LangChain for alert triage, sanctions screening, and entity resolution, routing risk scoring through deterministic Python services and cutting overall investigation latency 47%.
- Built **RAG pipelines** with LlamaIndex, OpenSearch, and Hugging Face embeddings using hybrid retrieval and cross encoder reranking across 14K+ compliance policy pages, achieving 94% retrieval precision and cutting guidance search time 67%.
- Fine tuned **Gemini Pro on Vertex AI** using 72K de identified compliance cases with PyTorch dataloaders and Hugging Face tokenizers, raising analyst agreement from 81% to 91% with MLflow quality gated evaluation blocking regressions.
- Implemented **guardrails** for PII redaction, regulatory output filtering, and hallucination detection against source records, achieving zero PII exposure incidents and reducing false regulatory citations from 8.2% to 0.6% in production.
- Served model inference via **FastAPI** with bounded concurrency, per request deadlines, and graduated retry budgets, sustaining 99.9% availability through canary releases with automated traffic shadowing and rollback during market hours.

Machine Learning Engineer | Jio Payments Bank | Hyderabad, India — Mar 2021 – Jul 2023

- Processed **35K card and UPI events per second** through Kafka and Flink streaming pipelines, transforming raw events into point in time features via PySpark for real time fraud scoring and reproducible offline model training.
- Trained **XGBoost and LightGBM ensembles** with graph based entity features and streaming rule triggers, improving confirmed fraud recall 14% and reducing false positive volume 19% while balancing detection quality against capacity.
- Built **FastAPI scoring APIs** with AsyncIO, Redis feature caches, and bulkhead isolation with deterministic thresholds and reason codes for explainable fraud decisions, cutting cascading service failures from 12 minutes to under 90 seconds.
- Automated model lifecycle with **MLflow, Airflow, Docker, and Kubernetes**, adding experiment tracking, canary rollouts, drift alarms, and one command rollback that cut release lead time from 3 weeks to 4 days across all models.
- Monitored scoring latency and fraud loss through **Prometheus and Grafana** dashboards, secured customer data with RBAC and encryption, and delivered Power BI reports on fraud loss trends and staffing projections to risk leadership.

Education

University of North Carolina at Charlotte — Aug 2023 – Dec 2024
M.S. in Data Science & Business Analytics (3.7 GPA) — Charlotte, NC

Research Publications

Melanoma Classification Using Deep Transfer Learning, IJDIIC, 1(1), 11–20, 2022
BOARD #322: An NSF FMRG Supported Exploratory Study of Prompting Large Language Models for a Conversational Manufacturing Education Platform, ASEE Annual Conference, 2025

Certifications

AWS Certified Generative AI Developer – Professional (AIP-C01), Amazon Web Services
Meta Back End Developer & Front End Developer Professional Certificates, Coursera